PROJECT SPECIFICATIONS – GROUP 21

AI-Powered Credit Default Risk Prediction System Using Machine Learning and Neural Network Techniques

GitHub Repository: https://github.com/REKNOS/AI-Powered-Credit-Default-Risk-Prediction-System

1. Project Overview

Credit default prediction is one of the most important challenges in the banking and financial industry. Financial institutions must accurately identify customers who are likely to default on loans or credit obligations to minimise financial losses and improve lending decisions.

Traditional credit-scoring systems often rely on predefined rules and statistical methods that may fail to capture complex relationships within large-scale financial datasets. Recent developments in Machine Learning (ML) and Deep Learning (DL) have significantly improved predictive performance; however, many of these models lack transparency and may unintentionally introduce bias against specific demographic groups.

This project aims to develop an AI-powered credit default risk prediction system that combines predictive accuracy, explainability, fairness, and ethical AI principles. Multiple Machine Learning and Neural Network models will be developed and compared using publicly available credit-risk datasets. Explainable AI (XAI), bias detection, and fairness evaluation techniques will be incorporated to ensure responsible and transparent decision-making.

2. Research Question / Problem Statement

Credit default prediction remains one of the most critical challenges in the banking and financial sector. Financial institutions rely on accurate risk assessment models to identify customers who are likely to default on credit obligations. Inaccurate predictions can lead to significant financial losses, poor lending decisions, increased operational risk, and reduced customer confidence. Traditional credit-scoring systems often depend on predefined statistical rules and may struggle to capture complex relationships hidden within large-scale financial datasets.

Recent advances in Machine Learning and Deep Learning have significantly improved predictive performance in credit-risk assessment. However, many high-performing models operate as black-box systems, making their decisions difficult to interpret and explain. In addition, predictive models may unintentionally introduce bias against particular demographic groups, raising concerns regarding fairness, transparency, and ethical AI deployment.

How can Machine Learning and Neural Network techniques be utilised to accurately predict credit default risk while maintaining fairness, transparency, explainability, and minimising algorithmic bias across diverse customer demographic groups? Furthermore, how can Explainable AI techniques and fairness evaluation frameworks be integrated into predictive models to support responsible and ethical lending decisions while analysing the trade-offs between predictive performance and fairness outcomes?

3. Project Objectives

• Analyse customer financial behaviour using structured credit datasets.
• Perform data preprocessing, cleaning, transformation, and feature engineering.
• Conduct Exploratory Data Analysis (EDA) and visualisation.
• Develop and compare multiple Machine Learning models and ANN models.
• Apply Explainable AI techniques using SHAP and LIME.
• Detect algorithmic bias and evaluate fairness across demographic groups.
• Investigate fairness-performance trade-offs.
• Develop a reproducible workflow using GitHub.
• Create a prototype decision-support system for credit risk assessment.

4. Goal / Target

The primary goal of this project is to develop an intelligent, explainable, and fairness-aware credit default prediction system capable of supporting financial institutions in making reliable lending decisions.

5. Key Performance Indicators (KPI)

Recent literature in credit-risk prediction reports that advanced Machine Learning techniques such as XGBoost, LightGBM, and Random Forest frequently achieve predictive accuracies above 90% and ROC-AUC scores between 0.90 and 0.95 on benchmark credit-risk datasets. Therefore, this project aims to achieve comparable or superior performance while maintaining fairness, transparency, explainability, and ethical AI standards.

KPI Category	Metric	Target Value
Predictive Performance	Accuracy	≥ 95%
Predictive Performance	Precision	≥ 90%
Predictive Performance	Recall	≥ 90%
Predictive Performance	F1-Score	≥ 90%
Predictive Performance	ROC-AUC	≥ 0.92
Predictive Performance	PR-AUC	≥ 0.90
Fairness	Demographic Parity Difference	≤ 0.10
Fairness	Equal Opportunity Difference	≤ 0.10
Fairness	Disparate Impact Ratio	0.80–1.25
Model Evaluation	Models Compared	Minimum 8 Models

6. Datasets and Resource Links

Dataset	Description	Source
Default of Credit Card Clients	Approximately 30,000 customer records including demographics, repayment behaviour and default status.	UCI Repository
Statlog German Credit Dataset	Customer financial profiles classified into good and bad credit-risk categories.	UCI Repository

7. Bias Detection and Fairness Evaluation

Bias Detection Techniques: Subgroup Performance Analysis, Group-wise Confusion Matrix Analysis, SHAP Feature Importance Comparison and Sensitive Attribute Impact Assessment.

Fairness Evaluation Metrics: Demographic Parity Difference, Statistical Parity Difference, Equal Opportunity Difference, Equalised Odds and Disparate Impact Ratio.

Bias Mitigation Techniques: Reweighing, Threshold Optimisation, Fairness-aware Post-processing and Bias Monitoring During Model Evaluation.

Tools: Fairlearn and IBM AIF360.

8. Ethical Considerations

This project will utilise publicly available credit-risk datasets obtained from the UCI Machine Learning Repository. All datasets are fully anonymised and do not contain personally identifiable information, ensuring that individual privacy is protected throughout the research process. Although the datasets contain sensitive financial attributes, no information can be linked to specific individuals, thereby minimising privacy and security risks.

The project will follow Responsible AI principles by incorporating fairness evaluation, bias detection, and explainability techniques throughout the model development lifecycle. Explainable AI methods such as SHAP and LIME will be used to improve transparency and accountability, while fairness assessment frameworks including Fairlearn and IBM AIF360 will be employed to identify and mitigate potential discriminatory outcomes. The proposed system is intended solely as a decision-support tool for financial institutions and will not replace human judgement in lending decisions. All project artefacts, including datasets, source code, documentation, and experimental results, will be securely maintained using GitHub version control and managed in accordance with University ethical guidelines and best practices for responsible AI development.

9. Tools and Technologies

Technology	Purpose

Python	Core Programming Language
Jupyter Notebook	Development Environment
Pandas	Data Processing
NumPy	Numerical Computation
Matplotlib	Data Visualisation
Seaborn	Statistical Visualisation
Scikit-learn	Machine Learning
XGBoost	Ensemble Learning
LightGBM	Gradient Boosting
TensorFlow / Keras	Deep Learning
SHAP	Explainable AI
LIME	Explainable AI
Fairlearn	Fairness Evaluation
IBM AIF360	Bias Detection & Mitigation
Git	Version Control
GitHub	Collaboration & Documentation